

SIMATS ENGINEERING

ASSESSMENT TOOL 2

**COURSE CODE: CSA1412
DESIGN**

COURSE NAME: COMPILER

COURSE OUTCOME COVERED

CO1: *Analyze syntax-directed translation method using synthesized and inherited attributes to generate a Parser Tree. (BL4)*

Assessment Tool Weightage: 20%

QUESTIONS: 5

**Duration :60 Mins
On Class
Total Marks:50**

Annexure A

CASE STUDY

Code of Conduct:

I Bhavya S(192524376) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Bhavya S

NAME : S. BHAVYA
REG NO : 192524376

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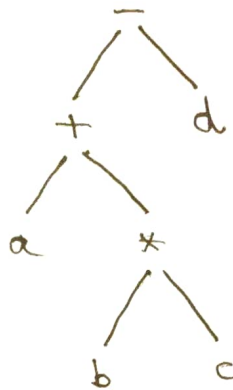
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1. Arithmetic Expression: $a + b * c - d$

i) Syntax Tree:

Since $*$ has higher precedence & $+$, $-$ are left associative.

$(a + (b * c)) - d$



ii) SDD:

$E \rightarrow E1 + T$
 $E \rightarrow E1 - T$
 $E \rightarrow T$
 $T \rightarrow T1 * F$
 $T \rightarrow F$
 $F \rightarrow id$



$E.val = E1.val + T.val$
 $E.val = E1.val - T.val$
 $E.val = T.val$
 $T.val = T1.val * F.val$
 $T.val = F.val$
 $F.val = id$

iii) Attribute Evaluation:

$$b * c = b.val \times c.val$$

$$a + b * c = a.val + (b * c).val$$

$$E.val = (a + b * c).val - d.val$$

for $a = 2, b = 3, c = 4, d = 5$

$$2 + (3 \times 4) - 5 = 9$$

iv) Conclusion:

The syntax tree ensures:

$$b * c \rightarrow a + (b * c) \rightarrow (a + (b * c)) - d$$

Thus, precedence & associativity are correctly represented.

2. S-Attributed SDD:

Grammar:

$$S \rightarrow E n$$

$$E \rightarrow E + T \mid E - T \mid F$$

$$T \rightarrow T * F \mid T / F \mid F$$

$$F \rightarrow (E) \mid \text{digit}$$

Input:

$$8 + 2 * 4 n$$

i) SDD:

$$E \rightarrow E_1 + T$$

$$E \rightarrow E_1 - T$$

$$E \rightarrow T$$

$$T \rightarrow T_1 * F$$

$$T \rightarrow T_1 / F$$

$$T \rightarrow F$$

$$F \rightarrow \text{digit}$$

$$F \rightarrow (E)$$



$$E.val = E_1.val + T.val$$

$$E.val = E_1.val - T.val$$

$$E.val = T.val$$

$$T.val = T_1.val * F.val$$

$$T.val = T_1.val / F.val$$

$$T.val = F.val$$

$$F.val = \text{digit} - \text{baseval}$$

$$F.val = E.val$$

ii) Translator:

$$t_1 = 2 * 4$$

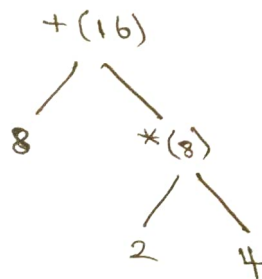
$$t_2 = 8 + t_1$$

$$\text{Result} = 16$$

iii) Stack Evaluation:

Stack	Action
8	Shift
F(8)	Reduce
T(8)	Reduce
E(8)	Reduce
E(8)+2	Shift
E(8)+T(2)*4	Shift / Reduce
E(8)+T(8)	$2 \times 4 = 8$
E(16)	$8 + 8 = 16$

iv) Syntax Tree:



v) Conclusion:

T handles *, so multiplication is performed before addition.

Final result = 16.

3. Attribute Rule Analysis:

Set	S-attributed	L-attributed	Circular Dependency
i) $A.s = B.i + C.i$	No	Yes	No
ii) $A.s = B.i + C.s ; D.i = A.i + B.s$	No	Yes	No
iii) $A.s = B.s + D.s$	Yes	Yes	No
iv) Given 4 rules	No	No	Yes

Set (iv) Circular Dependency

$$A.s \rightarrow B.i \rightarrow D.i \rightarrow A.s$$

Therefore, Set (iv) cannot be evaluated.

Conclusion:

Set (iii) is both S-attributed & L-attributed

Set (iv) contains a circular dependency & has no valid evaluation order.

4. Variable Declaration:

Grammar:

$$D \rightarrow TL$$

$$T \rightarrow \text{int} \mid \text{real}$$

$$L \rightarrow L, \text{id} \mid \text{id}$$

Input:

real p, q, r

1) SDD

$$D \rightarrow TL$$

$$L.in = T.type$$

$$T \rightarrow \text{int}$$

$$T.type = \text{int}$$

$$T \rightarrow \text{real}$$

$$T.type = \text{real}$$

$$L \rightarrow L, \text{id}$$

$$L.in = L.in$$

$$\text{id}.type = L.in$$

$$L \rightarrow \text{id}$$

$$\text{id}.type = L.in$$

ii) Attributes:

$\rightarrow T.type \rightarrow$ synthesized

$\rightarrow L.in \rightarrow$ inherited

$\rightarrow \text{id}.type \rightarrow$ identifier type.

iii) Evaluation:

$real \rightarrow T.type = real$

↓

$L.un = real$

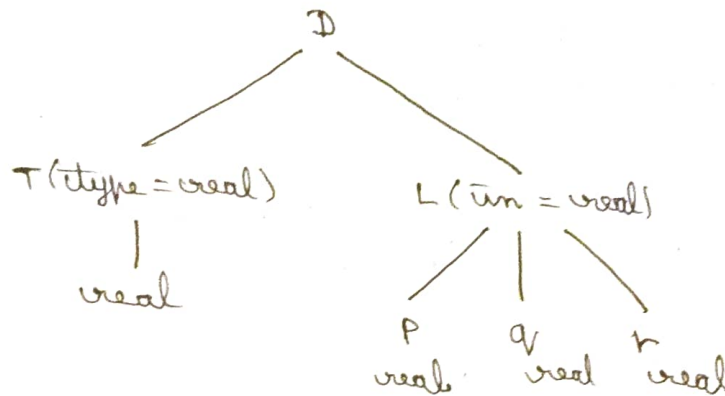
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$P.type = real$

$q.type = real$

$r.type = real$

iv) Annotated Tree:



v) Symbol Table:

Identifier	Type
P	real
q	real
r	real

Conclusion:

All identifiers correctly receive the type real.

5. Top-Down Translation

Input:

$3 * 4 + 6 * 7$

For top-down parsing, remove left recursion:

$E \rightarrow TE'$

$E' \rightarrow +TE' \mid \epsilon$

$T \rightarrow FT'$

$T' \rightarrow *FT' \mid \epsilon$

$F \rightarrow \text{digit} \mid (E)$

i) L-attributed SDD:

$$E \rightarrow TE'$$

$$E'.\text{inh} = T.\text{val}$$

$$E.\text{val} = E'.\text{syn}$$

$$E' \rightarrow +TE'1$$

$$E'1.\text{inh} = E'.\text{inh} + T.\text{val}$$

$$E'.\text{syn} = E'1.\text{syn}$$

$$E' \rightarrow \epsilon$$

$$E'.\text{syn} = E'.\text{inh}$$

$$T \rightarrow FT'$$

$$T'.\text{inh} = F.\text{val}$$

$$T.\text{val} = T'.\text{syn}$$

$$T' \rightarrow *FT'1$$

$$T'1.\text{inh} = T'.\text{inh} * F.\text{val}$$

$$T'.\text{syn} = T'1.\text{syn}$$

$$T' \rightarrow \epsilon$$

$$T'.\text{syn} = T'.\text{inh}$$

$$F \rightarrow \text{digit}$$

$$F.\text{val} = \text{digit}.\text{lexval}$$

iii) Attribute Evaluation

$$3 \rightarrow F.\text{val} = 3$$

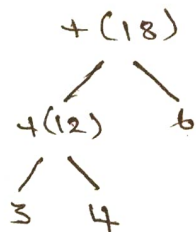
$$3 \times 4 \rightarrow 12$$

$$12 + 6 \rightarrow 18$$

iii) Final Result

$$3 \times 4 + 6 = 18$$

iv) Annotated Tree:



Conclusion:

Multiplication is performed first because T handles $*$

Final Answer = 18.